

AI Academic Mentor

Personalized Learning & Study Assistant

Initial Project Plan — CSE4104-7C-T05 | CSE 4104 | NUBTK | 25 May 2026

Project Summary

14	6	4	8
Weeks	Phases	Team Members	Core Features

Team Members

Role	Name	Student ID
Team Leader	Shadman Sadequeen Priyo	11230121158
Frontend Developer	Musabbir Hossain Chayon	11230121168
Backend Developer	Sk. Nadirul Haque Nadir	11230121155
AI Integration Lead	Shafin Kabir	11230121175

14-Week Development Timeline

Week 1–2	Planning	Team Formation & Project Proposal	Completed
Deliverable:	GitHub repo + Project Proposal PDF - Form team and assign roles: Leader, Frontend Dev, Backend Dev, AI Integration Lead - Create GitHub repo with branch structure: main / development / frontend / backend / ai-integration - Finalize technology stack: React.js, Node.js, MongoDB, Gemini/OpenAI API - Submit Week 2 Project Proposal document		
Week 3–5	Design	SRS, Architecture & UI/UX Design	Upcoming
Deliverable:	SRS document · ER diagram · Figma wireframes - Week 3: Write Software Requirements Specification (SRS) + use case diagrams - Week 4: Design system architecture diagram, ER diagram, and REST API structure - Week 5: Create Figma wireframes for all major pages (Login, Dashboard, Chatbot, Quiz, Summarizer, Planner)		
Week 6–7	Development	Backend & Frontend Development	Upcoming
Deliverable:	Auth APIs · Database setup · All UI pages connected to backend - Week 6: Backend — JWT auth system, bcrypt password hashing, REST API endpoints, MongoDB Atlas setup - Week 6: Backend — Database schemas for users, quizzes, study plans, and progress data - Week 7: Frontend — Build all UI pages in React.js + Tailwind CSS - Week 7: Frontend — Connect all pages to backend REST APIs and test data flow		
Week 8–9	AI Integration	AI Integration & Feature Completion	Upcoming
Deliverable:	Chatbot · Quiz Generator · Summarizer · Study Planner · Dashboard Week 8: Integrate AI Chatbot (Gemini/OpenAI) with conversation history per student session		

- Week 8: Build AI Quiz Generator — MCQ + short-answer formats, score tracking in database
- Week 8: Implement AI Note Summarizer — paste or upload notes, generate concise summaries
- Week 9: Build Personalized Study Planner — AI generates structured day-by-day plans
- Week 9: Complete Progress Dashboard with charts, topic-wise analytics, and engagement metrics
- Week 9: Polish UI, add in-app notification system and Admin Panel

Week 10–11	QA & Deployment	Testing, Debugging & Deployment	Upcoming
Deliverable:	Testing report · Live deployment on Vercel + Render/Railway • Week 10: Write test cases, run unit + integration + user acceptance testing • Week 10: Document all bugs and fixes in a testing report • Week 11: Deploy frontend to Vercel • Week 11: Deploy backend API to Render / Railway • Week 11: Verify live deployment and update README.md with deployment link		
Week 12–14	Presentation	Documentation, Prep & Final Viva	Upcoming
Deliverable:	Final report · User manual · API docs · Live demo • Week 12: Write final project report, user manual, and API documentation • Week 13: Prepare presentation slides and rehearse live demonstration • Week 14: Final presentation, live demo, and Q&A; viva evaluation		

Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS	User interface and responsive design
Backend	Node.js + Express.js	REST API, business logic, authentication
Database	MongoDB + MongoDB Atlas	Flexible data storage and cloud hosting
AI Integration	Gemini API / OpenAI API	Chatbot, quiz generation, summarization
Authentication	JWT + bcrypt	Secure login and session management
Frontend Deploy	Vercel	Fast, free frontend hosting
Backend Deploy	Render / Railway	Free backend API hosting
Version Control	GitHub	Code management and team collaboration